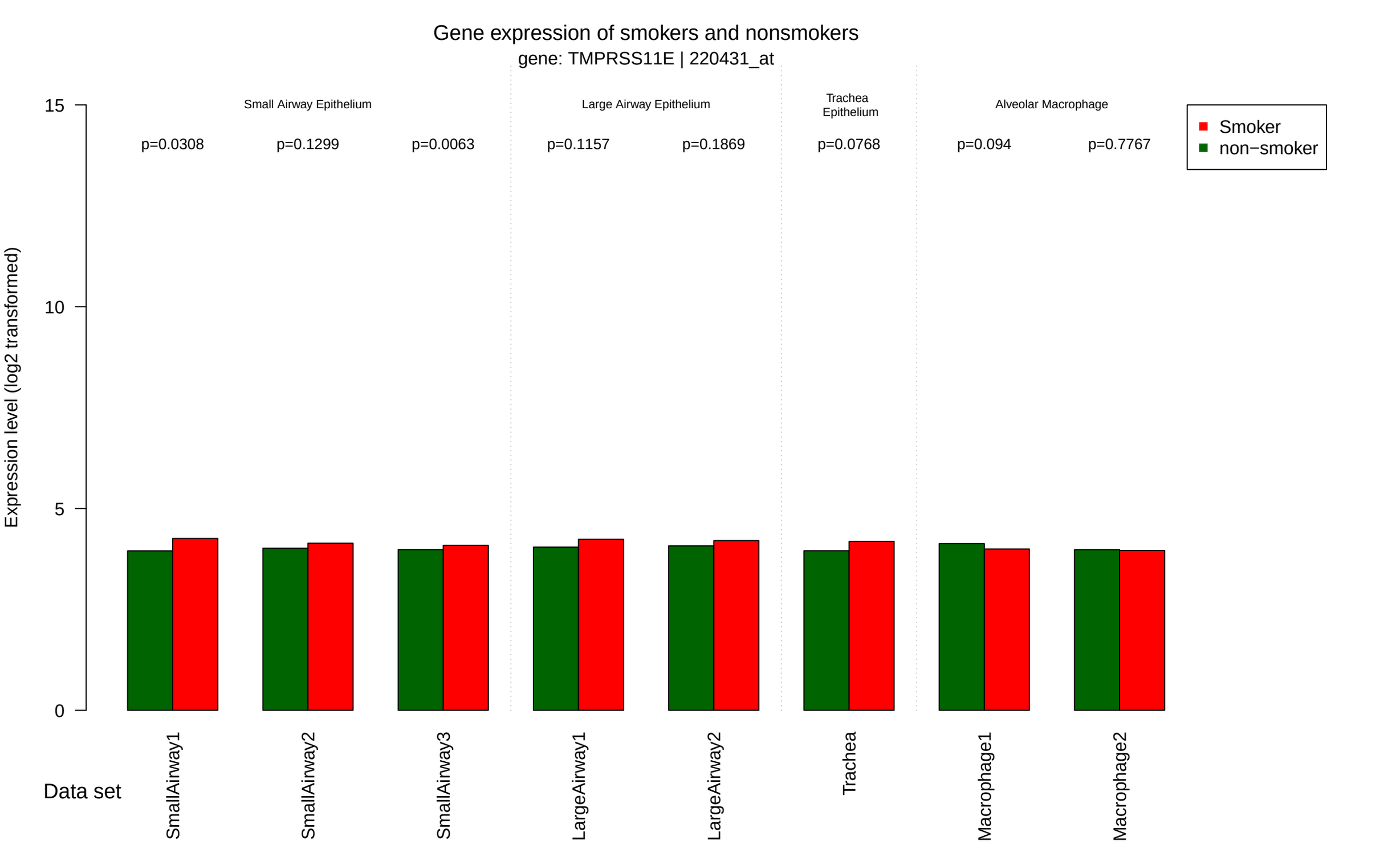
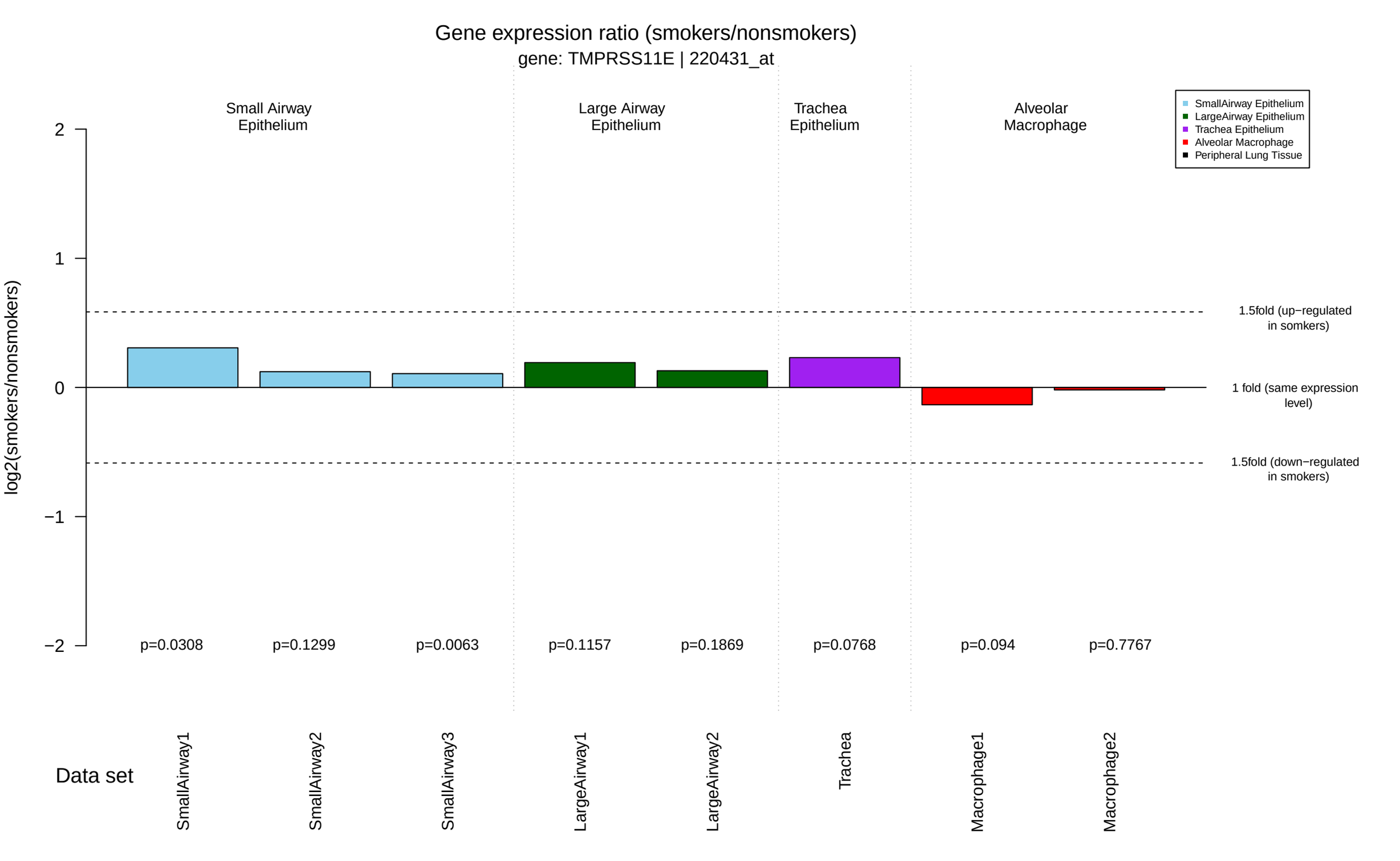
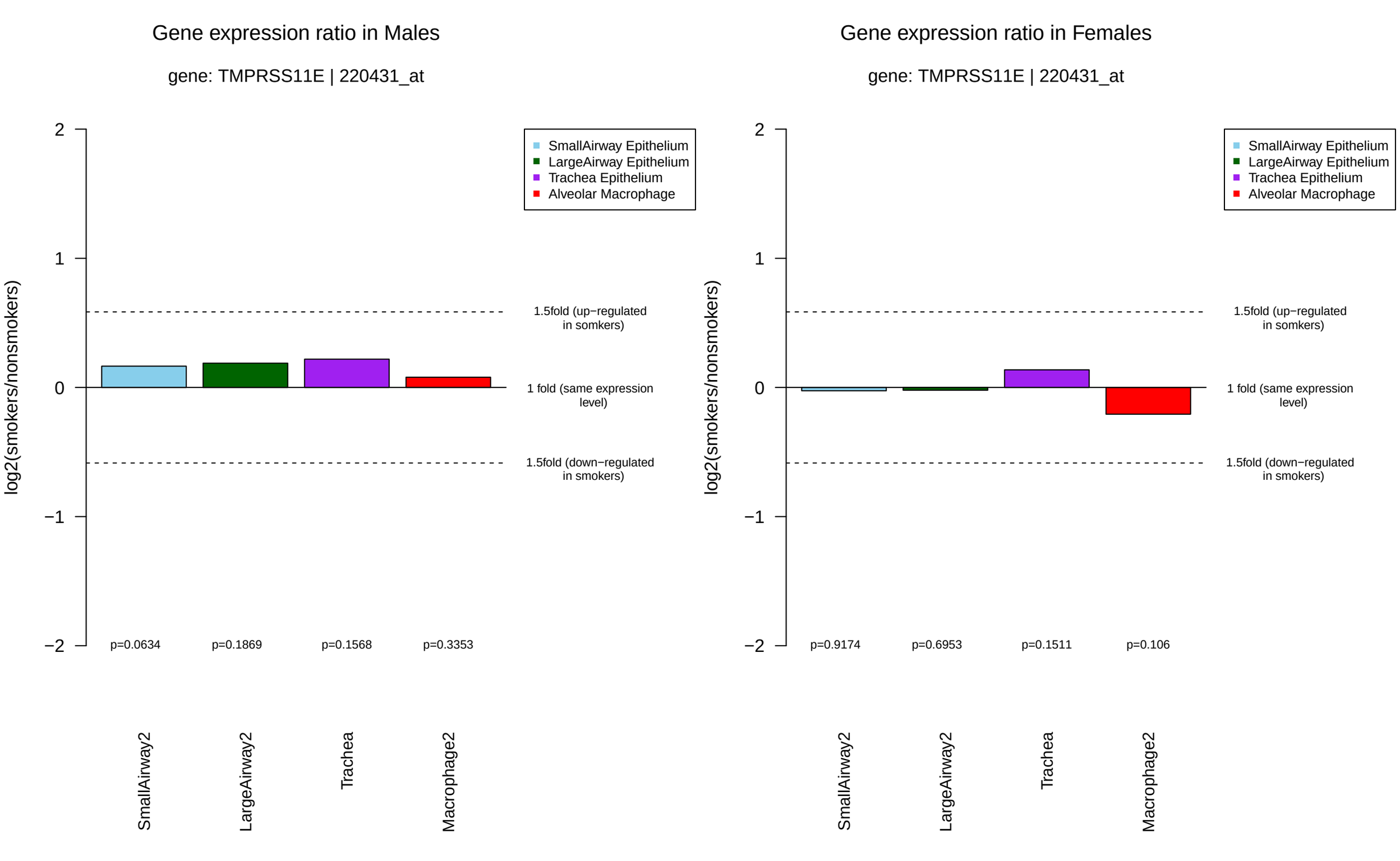
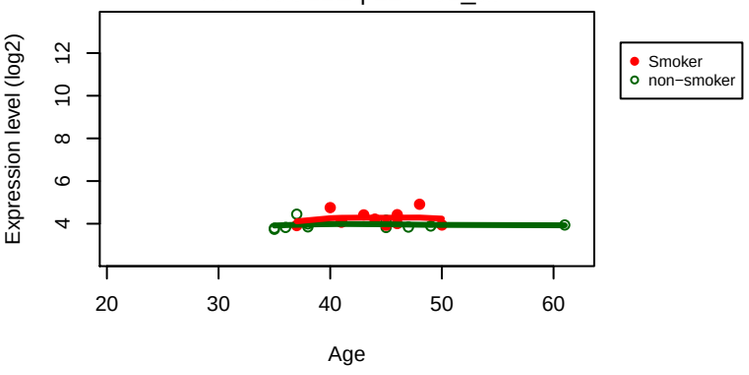
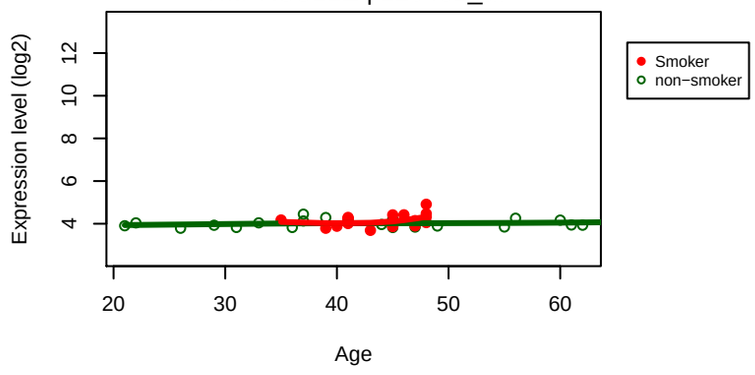
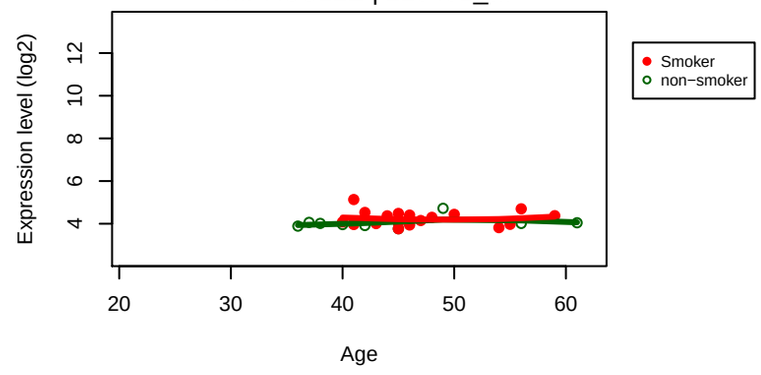

Gene expression of smokers and nonsmokers
gene: TMPRSS11E | 220431_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0308
p=0.1299
p=0.0063
p=0.1157
p=0.1869
p=0.0768
p=0.094
p=0.7767
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

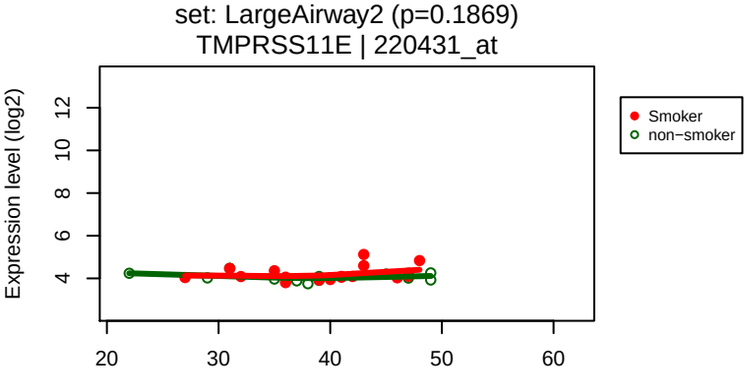

Gene expression ratio (smokers/nonsmokers)
gene: TMPRSS11E | 220431_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0308
p=0.1299
p=0.0063
p=0.1157
p=0.1869
p=0.0768
p=0.094
p=0.7767
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

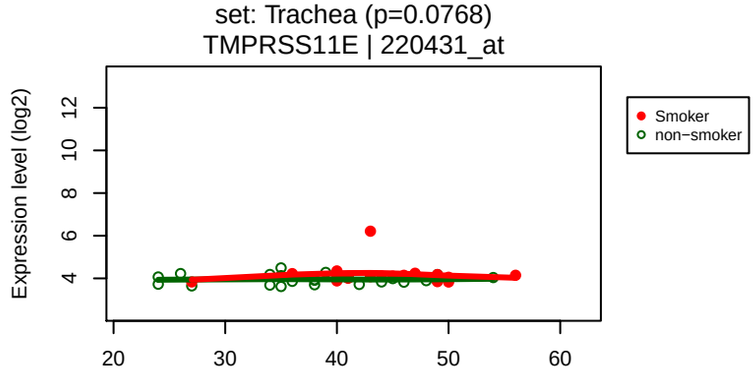

Gene expression ratio in Males
gene: TMPRSS11E | 220431_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0634
p=0.1869
p=0.1568
p=0.3353
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMPRSS11E | 220431_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9174
p=0.6953
p=0.1511
p=0.106
SmallAirway2
LargeAirway2
Trachea
Macrophage2

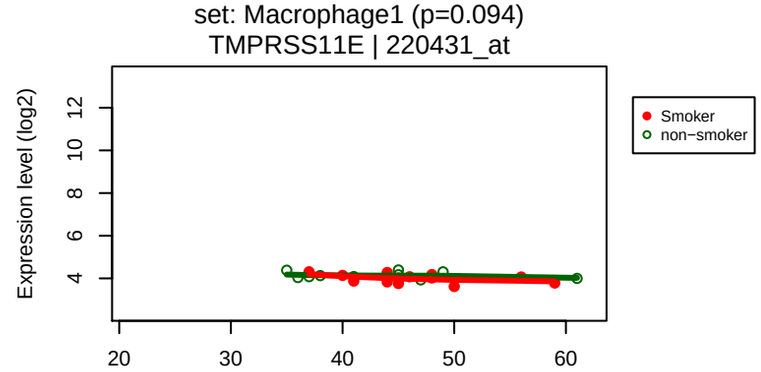

set: SmallAirway1 (p=0.0308)
TMPRSS11E | 220431_at
Expression level (log2)
Age
Smoker
non−smoker

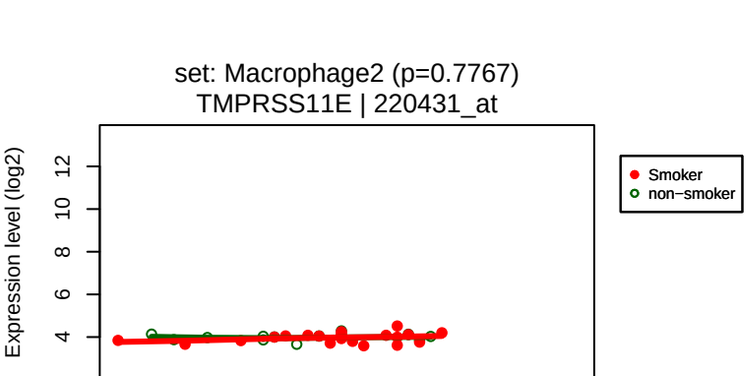

set: SmallAirway2 (p=0.1299)
TMPRSS11E | 220431_at
Expression level (log2)
Age
Smoker
non−smoker

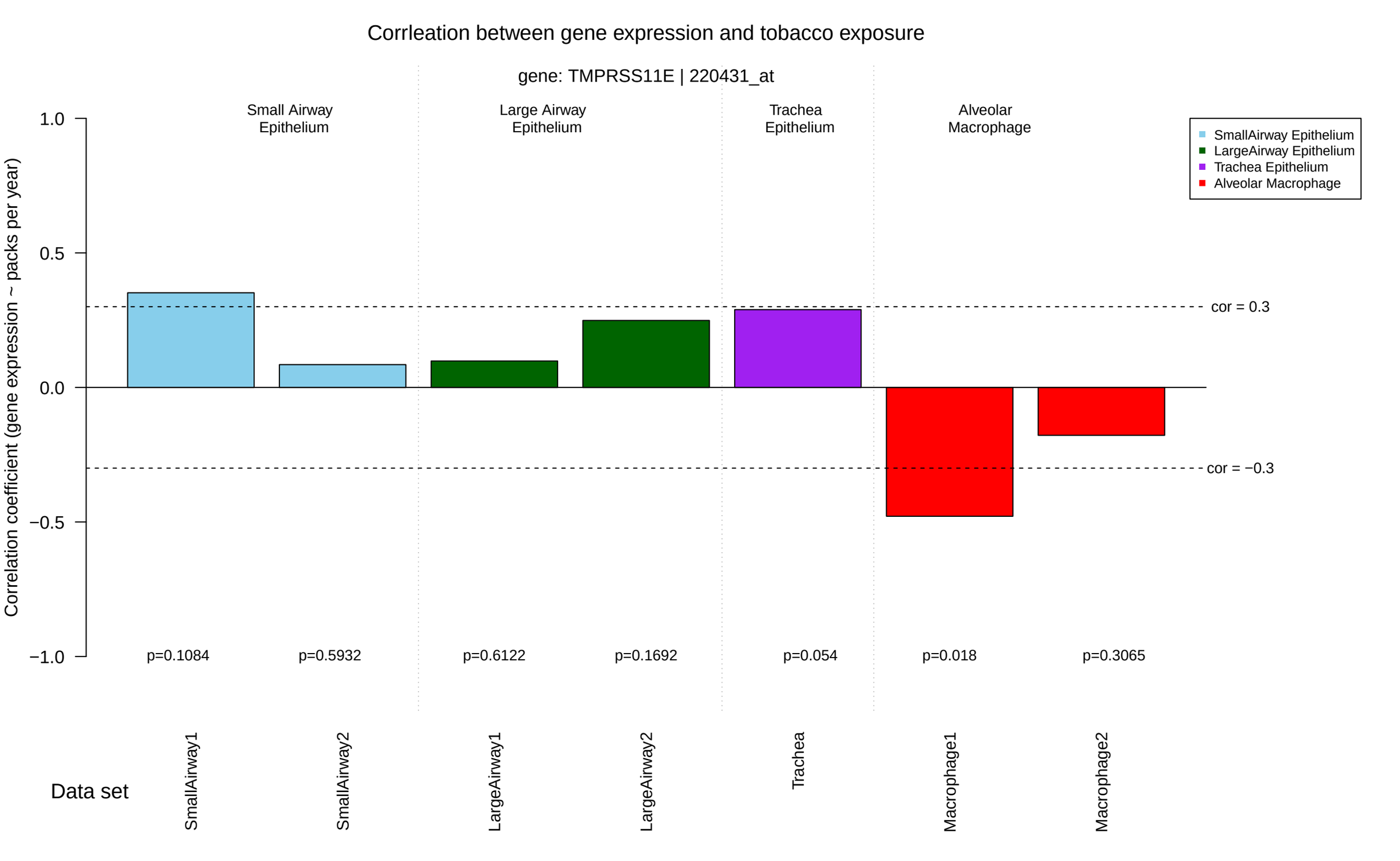

set: LargeAirway1 (p=0.1157)
TMPRSS11E | 220431_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1869)
TMPRSS11E | 220431_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0768)
TMPRSS11E | 220431_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.094)
TMPRSS11E | 220431_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7767)
TMPRSS11E | 220431_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMPRSS11E | 220431_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1084
p=0.5932
p=0.6122
p=0.1692
p=0.054
p=0.018
p=0.3065
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2